

Atul Kumar Rai

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SUMMARY

Software engineer with 7+ years of experience designing and operating distributed systems across PlayStation, AWS, and enterprise environments. Led ownership of critical session management and GraphQL federation services, built platform-wide performance testing frameworks, and drove cross-team technical initiatives across US, Japan, and India. Focused on systems design, service reliability, observability, and performance optimization.

EXPERIENCE

Software Engineer II, Sony Interactive Entertainment LLC, San Mateo, California, United States 09/11/2023 – Present

- Reduced Game Media Service (GMS) video transcoding egress costs by ~\$500K/year by restricting instantaneous transcoding to active CMG PlayStation App members, cutting average daily egress cost from \$5,578 to \$4,207.
- Diagnosed and resolved a critical memory leak in arkham-session that had persisted for over two years, performing deep root cause analysis and presenting findings to North America Services leadership.
- Designed and built the IsoPerf performance testing framework end-to-end for Arkham GraphQL (Core and Profile subgraphs), including test environments, mock services, Helm charts, k6 scripts, and Grafana dashboards — adopted platform-wide for data-driven feature rollout decisions.
- Assumed primary ownership of off-console session management and proxy services, leading parallel cross-team initiatives including PlayStation.com integration, Family mobile app onboarding, and PlayStation mobile app onboarding.
- Designed and delivered the FriendsWhoPlayed API serving multiple client teams with performance validation through IsoPerf testing against required latency thresholds.
- Architected mobile session lifecycle management for PlayStation mobile app and Family mobile app, designing server-side opaque session handles, cross-app SSO via multi-tenant DynamoDB, and device-bound security aligned with OAuth 2.0 best practices.
- Led AWS CloudWatch Synthetic Canaries initiative for sandbox testing from POC through team-wide adoption across US, Japan, and India teams; built sandbox management features including soft deletion, pagination, and the searchSandboxes API.
- Expanded platform observability across Arkham GraphQL by adding service metrics and Grafana dashboards, integrating LTIE/LTIP resiliency frameworks with the perf-ci team, and leading on-call incident response with end-to-end root cause analysis.

Software Development Engineer, Amazon Web Services, Santa Clara, California, United States 09/19/2022 – 06/26/2023

- Designed an anti-entropy reconciliation system to detect and repair data inconsistencies from dropped notifications, targeting a 12% reduction in monthly escalations.
- Optimized AWS Lambda functions and CloudWatch resource management, resolving critical bottlenecks and reducing average API response time.
Built event-driven microservices using Lambda, SNS, SQS, and DynamoDB with CloudFormation IaC, improving deployment consistency and customer experience.
- Automated custom report generation using API Gateway, S3, and iText, streamlining customer information delivery.

Senior Systems Engineer, Infosys Limited, Pune, India 06/12/2017 – 04/30/2021

- Designed and developed scalable RESTful APIs in C# for a Fortune 500 insurance client, integrating Federal, State, and company regulatory requirements into service communication layers.
- Optimized SQL queries for high-volume data processing, improving retrieval performance and storage efficiency across production databases.
- Performed root cause analysis using data-driven techniques, reducing mean resolution time by 20% and improving application stability.
- Automated recurring workflows with Python scripts, reducing manual intervention and accelerating key operational processes.
- Mentored engineers across geographically distributed teams on SOLID principles, OOP design, and engineering best practices.

PROJECTS

Stream Processing with Kafka and Samza

- Developed a backend process for an Uber-like application by leveraging Kafka and Samza, enabling efficient handling of driver, rider, and advertisement data through the creation of topics in Kafka.
- Leveraged Samza to analyze Kafka streams, facilitating optimal rider-driver matching by considering preferences, rating, salary, and location, ensuring an efficient and personalized experience.

Microservice Multi-Cloud Deployment

- Deployed 3 microservices for a WeChat-inspired application across Kubernetes clusters on AWS, GCP, and Azure, ensuring platform flexibility and mitigating vendor lock-in risks.
- Created and deployed Helm charts for simplified management, provisioning, updates, and rollbacks, reducing deployment time and enhancing release management in the Kubernetes environment.

EDUCATION

Masters, Information Systems Management
Carnegie Mellon University

05/10/2021 – 08/18/2022
GPA: 3.93

Bachelor of Technology, Electrical and Electronics Engineering
SRM University

07/15/2013 – 05/10/2017
GPA: 3.95

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Certified Kubernetes Application Developer (CKAD) — In Progress
- Apollo GraphQL Workshop — Sony Interactive Entertainment, October 2023
- GDFT Team 6 AI Workshop — Selected among top 15; hands-on training in LangChain, RAG architectures, and LLM integration, 2025
- GMX Mentorship Program — Mentored apprentice engineers through AnitaB.org, 2024–2025

SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C#, Shell Script

Frameworks & Libraries: Apollo Server, Apollo Federation, Apollo Router, Express.js, Node.js, GraphQL, Spring Boot, JPA, Hibernate, Swagger/OpenAPI

Cloud & Infrastructure: Kubernetes, Helm, Terraform, Docker, AWS (Lambda, DynamoDB, SQS, SNS, S3, CloudFormation, CloudWatch)

Databases & Messaging: Cassandra, Redis, MongoDB, Kafka, Kinesis, PostgreSQL, MySQL

Testing & Performance: k6, Jest, JUnit, Mockito, Supertest, Clinic.js, IsoPerf

Build & Observability: Datadog, Grafana, Prometheus, Splunk, Jenkins, Lerna, GitHub, Maven, Jira